

Bo Shao

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Research Interests

Methodological Areas: Data-Driven Modeling and Analysis, Operations Research, System Analytics and Optimization.
Application Areas: Production and Operations Systems, Logistics Networks, Supply Chain Management.

Education

North Carolina State University

Aug. 2025 – Present

Doctor of Philosophy in Industrial Engineering

Master of Operations Research (En Route)

- GPA: 4.0/4.0

University of Wisconsin - Madison

Sep. 2023 – May 2025

Master of Science in Industrial Engineering: Systems Engineering and Analytics

- GPA: 3.8/4.0
- Coursework: Applied Temporal Data Analytics, Machine Learning, Production Systems Control, OR-Probabilistic Modeling, Supply Chain Modeling, Decision Making in Health Care, Stochastic Processes, Linear Programming.

Fuzhou University

Sep. 2017 – Jun. 2021

Bachelor of Engineering in Logistics Engineering

- GPA: 3.53/4.0 (**Top 10%**).
- Coursework: Statistics, System Simulation, Operations Research, Traffic Engineering, Supply Chain Management.

Publications

- **Bo Shao** & L. Zhang. (2026). Beyond Prediction: AI Agents in the Next Era of Environmental Early Warning. *Integrated Environmental Assessment and Management*, 22(4), 1338–1340. DOI:10.1093/inteam/vjag077.
- **Bo Shao**, Xiaoli Su, Xin Li & Xingxuan Zhuo. (2025). Mixed-Frequency Data-Driven Forecasting Port Throughput: A Novel Attention-DeepAR-MIDAS Model. *International Journal of Shipping and Transport Logistics*, 20(3), 338–358. DOI:10.1504/IJSTL.2025.147408 [🔗](#) [📄 \[PDF\]](#)
- Xiaoli Su, Lijun Zeng, **Bo Shao** & Binlong Lin. (2025). Data-Driven Optimization for Production Planning with Multiple Demand Features. *Kybernetes*, 54(1), 110–133. DOI:10.1108/K-04-2023-0690 [🔗](#) [📄 \[PDF\]](#)
- Xin Li, Yongshi Hu, Xiaoli Su & **Bo Shao**. (2023). A Data-Driven Newsvendor Problem with Shifting Demand: A Deep Autoregressive Model with Attention Mechanism. *Journal of Engineering Science & Technology Review*, 16(3), 74–83. DOI:10.25103/jestr.163.10 [🔗](#) [📄 \[PDF\]](#)
- Jianping Zheng & **Bo Shao**. (2022). Throughput Properties and Clustering Analysis of Coastal Ports in Mainland China: An Analysis Method by Time Series Complex Network. *DYNA-Ingeniería e Industria*, 97(4), 398–405. DOI:10.6036/10537 [🔗](#) [📄 \[PDF\]](#)

Project & Research Experience

Data-Driven Modeling and Optimization for Production and Operations Systems

Jun. 2023 – Present

Advisors: Prof. Chenghu Yang (Fuzhou, CN) and Prof. Xin Ma (Melbourne, AU).

- Conducted an extensive literature review on the data-driven newsvendor problem, the production planning problem, demand forecasting with machine learning, and the high-dimensional mixed-frequency approach.
- Conducted numerical experiments to evaluate the performance of the proposed models, analyzing the impact of demand granularity, demand censoring, and varying levels of risk aversion on order decisions under different conditions.
- Collected and cleaned data for the empirical study and validated the practical effectiveness of the data-driven newsvendor model integrated with the Attention-DeepAR method. [📄](#)

- o Participated in a data-driven optimization model development, applying the mixed data sampling (MIDAS) technique to address a production problem with multiple features. 
- o **Outcome:** Published two academic papers in peer-reviewed journals, with another paper currently under review.

Reshoring U.S. Manufacturing: Analysis of Risks and Capabilities

Jan. 2024 – May 2025

Advisor: Prof. David Ding.

Madison, US

- o Authored a research report synthesizing findings from both academic and industry sources, and delivered an oral presentation at the 2024 MBAA International Conference in Chicago, IL, engaging with experts from different fields.
- o **Outcome:** Bo Shao, David Ding & Yanhui Huang. (2024). *The challenges for the U.S. manufacturing to reshore supply chain*. In 2024 MBAA International Conference, Chicago, IL, United States, April 11, 2024. 

Data-Driven Analysis and Forecasting for Port Throughput

Jul. 2021 – May 2023

Advisors: Prof. Chenghu Yang and Prof. Xingxuan Zhuo.

Fuzhou, CN

- o Collected and constructed a shipping dataset consisting of six mixed-frequency variables spanning 180 months.
- o Participated in Attention-DeepAR-MIDAS model development and conducted numerical experiments to assess performance in handling multiple mixed-frequency features. 
- o Conducted time series network analysis of China's 25 largest coastal ports, focusing on hierarchical structures and clustering features of port networks, and applied GIS to visualize their topological characteristics. 
- o **Outcome:** Published two academic papers in peer-reviewed journals.

Working & Teaching Experience

INFORMS Student Chapter, North Carolina State University

Aug. 2026 – Present

President

Raleigh, NC

- o Serve as President of the NC State INFORMS Student Chapter, coordinating chapter activities, student engagement, and academic and professional events.

I SY E 615 – Production Systems Control, University of Wisconsin–Madison

Fall 2024

Teaching Assistant, Department of Industrial and Systems Engineering

Madison, WI

- o Supported course instruction by holding office hours, assisting students with course materials and problem solving, and grading homework and quizzes.

Logistics Research Center of Fuzhou University

Jul. 2021 – Aug. 2023

Research Assistant (Full-Time)

Fuzhou, CN

- o Conducted research on logistics, supply chain management, data-driven modeling, and optimization.
- o Collaborated on academic research projects, manuscript preparation, data analysis, and research proposals.

Honors & Awards

- o 2026 Edward P. Fitts Graduate Fellowship, NC State ISE *Aug. 2026*
- o 2025 Edward P. Fitts Graduate Fellowship, NC State ISE *Aug. 2025*
- o 2021 Summa Cum Laude Graduate of Fuzhou University (Top 5%) *Jun. 2021*
- o 2021 Sci-Tech Innovation Individual Scholarship (Top 3%) *Apr. 2021*
- o Academic Merit Scholarship, First Tier (Top 5%) *Oct. 2020*
- o 2020 SUNSHINE Individual Scholarship (Top 3%) *Apr. 2020*
- o Academic Merit Scholarship, Third Tier (Top 15%) *Oct. 2019*
- o Second Prize in the National Contest on Logistics Design (NCOLD) *Apr. 2019*

Skills & Certificates

Programming Languages: C++, Python, R, SQL.

Software and Tools: ArcGIS, GeoDa, Tableau, Origin, LaTeX, AutoCAD, Arena, Stata, SPSS, MATLAB, Mathematica.

Certificates: Lean Six Sigma Green Belt (AIGPE, 2024); Chinese Certified Professional Logistician (CPL).

Hobbies: Hiking, Climbing, Cooking.